Eigenfaces with NumPy

A Manual Principal Component Analysis Implementation for Face Representation

Abstract

This report presents a NumPy implementation of the Eigenfaces method for representing facial images in a low-dimensional linear subspace. The supplied ORL face dataset contains 400 grayscale images from 40 individuals. Each image is resized to 100 by 100 pixels and converted into a vector of 10,000 intensity values. Principal component analysis is implemented through mean subtraction, eigendecomposition of a sample-space Gram matrix, and transformation of its eigenvectors into the original pixel space. The resulting model retains 25 orthonormal eigenfaces and represents the dataset using a coefficient matrix of size 400 by 25. Numerical analysis of the supplied data shows that these components preserve 74.03% of the total centered variance. Reconstruction with the same basis produces a root mean squared error of 19.92 intensity units on the fitted dataset. The mean face and the eigenface images provide qualitative evidence of the extracted facial structure, while numerical residuals support the correctness of the implementation. The experiment demonstrates dimensionality reduction and feature extraction; classification accuracy and generalization to unseen images are outside the implemented evaluation.

Keywords: Eigenfaces; principal component analysis; NumPy; face representation; dimensionality reduction.

1 Introduction

A grayscale face image is naturally represented by a high-dimensional vector whose entries correspond to pixel intensities. Direct use of this representation preserves all recorded information but also retains correlated and redundant variation. A compact representation is therefore useful for examining the dominant structure of a collection of facial images and for constructing features for subsequent analysis.

Principal component analysis (PCA) identifies orthogonal directions along which centered observations exhibit the greatest variance. In the Eigenfaces approach, these directions are reshaped into images. Each eigenface describes a pattern of variation across the dataset, and each observed face can be expressed approximately as the mean face plus a weighted combination of selected eigenfaces.

The objective of this project is to make the complete computational procedure explicit using NumPy linear algebra. The implementation loads and labels images, computes the mean face, extracts a basis of 25 components, visualizes the basis, and projects the original observations into the reduced space. The analysis below explains these stages and evaluates their numerical behavior on the supplied dataset.

2 Dataset and Preprocessing

The supplied dataset is organized into 40 subject directories, named s1 through s40, with ten PGM images in each directory. Inspection confirms that all 400 images are readable grayscale images with a width of 92 pixels and a height of 112 pixels. The subject directory identifies the person associated with an image; it is used to assign a numeric label during loading.

Property	Configuration
Number of subjects	40
Images per subject	10
Total number of images, N	400
Original image dimensions	92 pixels wide by 112 pixels high
Processed image dimensions	100 pixels wide by 100 pixels high
Features per processed image, D	10,000
Retained principal components, K	25
Numerical data type	NumPy float64

2.1 Image loading and vectorization

The load_images function visits the subject directories in lexicographic order and assigns consecutive numeric labels. Consequently, directory names such as s10 precede s2. Image files within a directory follow the order returned by the file system. This ordering affects row positions and label assignments, but the mathematical PCA subspace is invariant to a consistent permutation of the observations.

Each image is converted to grayscale using Pillow and resized to 100 by 100 pixels. Because the original images are rectangular, this operation changes their aspect ratio. The resized image is converted to a float64 array and flattened into a vector. Collecting these vectors as rows produces the data matrix X with dimensions 400 by 10,000. The label vector and a label-to-directory mapping are returned with this matrix.

2.2 Intensity scale and centering

The implementation retains pixel intensities on their original 8-bit scale. It does not divide pixel values by 255 or standardize each pixel to unit variance. In the resized dataset, the observed intensities range from 0 to 254. Conversion to floating point supports subsequent arithmetic without changing this intensity scale.

Mean subtraction is performed as part of PCA. It removes the average value of each pixel across all observations, so the extracted directions describe deviations from the mean face. This centering operation is distinct from rescaling image intensities. The eigenfaces themselves are later normalized to unit Euclidean length.

3 Mathematical Formulation

3.1 Mean face and centered observations

Let X be the N by D data matrix, with one row x_i for each image. Here N = 400 and D = 10,000. Denote the mean image vector by m and the centered matrix by A. The mean is computed independently for each pixel:

$$m = (x_1 + x_2 + ... + x_N)/N, A_{i,:} = x_i - m$$

The compute_mean_face function implements the column-wise mean using np.mean(X, axis=0). Reshaping m to 100 by 100 pixels yields the mean-face image shown in Figure 1.

3.2 Eigensolution in sample space

The feature-space covariance matrix may be written as $C = A^T A/N$. Its dimensions are 10,000 by 10,000. The implementation instead forms the smaller, symmetric Gram matrix L in sample space:

$$L = AA^T, Lv_j = l_j v_j$$

The matrix L has dimensions 400 by 400. NumPy computes its eigenvalues and eigenvectors with np.linalg.eigh, which is appropriate for a real symmetric matrix. The eigenpairs are sorted by descending eigenvalue, placing directions with the largest variance first. Because the observations are centered, the maximum possible rank is N - 1 = 399.

The nonzero eigenvalues of L are N times the corresponding eigenvalues of C. The missing factor 1/N changes the eigenvalue scale but does not change the eigenvectors, their ordering, or explained-variance ratios. This distinction is necessary when interpreting L as an efficient substitute for the covariance eigensolution.

3.3 Eigenface construction and normalization

A sample-space eigenvector v_j is mapped to the pixel space and normalized. The retained columns form the D by K matrix U:

$$b_j = A^T v_j, u_j = b_j / ||b_j||_2, U = [u_1, ..., u_K]$$

Each column of U contains 10,000 values and can be displayed as an eigenface. Normalization produces unit-length directions. Components associated with zero or numerically negligible eigenvalues require special handling in a generalized implementation; the retained 25 components in this experiment have valid, nonzero norms.

3.4 Projection and approximate reconstruction

$$Z = AU, X_{approx} = ZU^T + m$$

The project_faces function produces Z, the N by K matrix of eigenface coefficients. Approximate reconstruction adds the mean face to the weighted basis expansion. Reconstruction is used here as an additional numerical evaluation; the original script computes the projection but does not include a reconstruction display or classifier.

4 Quantitative Results

The original loading, mean, eigenface, and projection functions were evaluated on all 400 supplied images. Additional diagnostics were computed from the same centered data and basis without modifying the model. All reported variance and reconstruction results describe the dataset used to fit PCA; no training and test partition was used.

Measure	**Measured result**
Input matrix X	400 by 10,000
Eigenface matrix U	10,000 by 25
Projected matrix Z	400 by 25
Variance retained by 25 components	74.03%
Variance excluded by 25 components	25.97%
Reconstruction mean squared error	396.88 intensity units squared
Reconstruction root mean squared error	19.92 intensity units
Maximum absolute centered column mean	1.00×10^{-13}
Maximum absolute entry of $U^TU - I$	5.55×10^{-15}
Maximum relative eigenpair residual	2.08×10^{-14}

4.1 Variance retention and dimensionality

The first component accounts for 17.88% of the total centered variance. The first five components retain 48.78%, and the first ten retain 60.86%. Retaining 25 components increases this proportion to 74.03%. Under the measured spectrum, 40 components are required to reach 80% variance, 101 to reach 90%, and 177 to reach 95%. Thus, 25 components provide substantial dimensionality reduction while discarding a measurable portion of the observed variation.

Each image is represented by 25 coefficients instead of 10,000 pixel values, corresponding to a 400-fold reduction in feature count. This is a reduction in the dimensionality of each representation; the complete model also requires the mean vector and the eigenface basis to interpret or reconstruct those coefficients.

4.2 Reconstruction and numerical consistency

Reconstruction error was calculated as the average squared difference over all 4,000,000 entries of X. Its square root, 19.92, expresses error in the same intensity units as the input. This aggregate measure describes reconstruction of fitted images and does not establish recognition performance or image quality on unseen subjects.

The centered-column residual is close to floating-point precision, and U^TU is numerically close to the identity. The relative eigenpair residual was measured as $\|A^TAu_j - l_ju_j\|_2/l_j$ for each retained unit vector. These diagnostics support the expected centering, orthonormality, and eigenvector relationships.

5 Visual Results



Figure 1. Mean face obtained by averaging the 400 processed images. The central facial structure remains visible, while variation between individuals and images produces a smooth appearance.

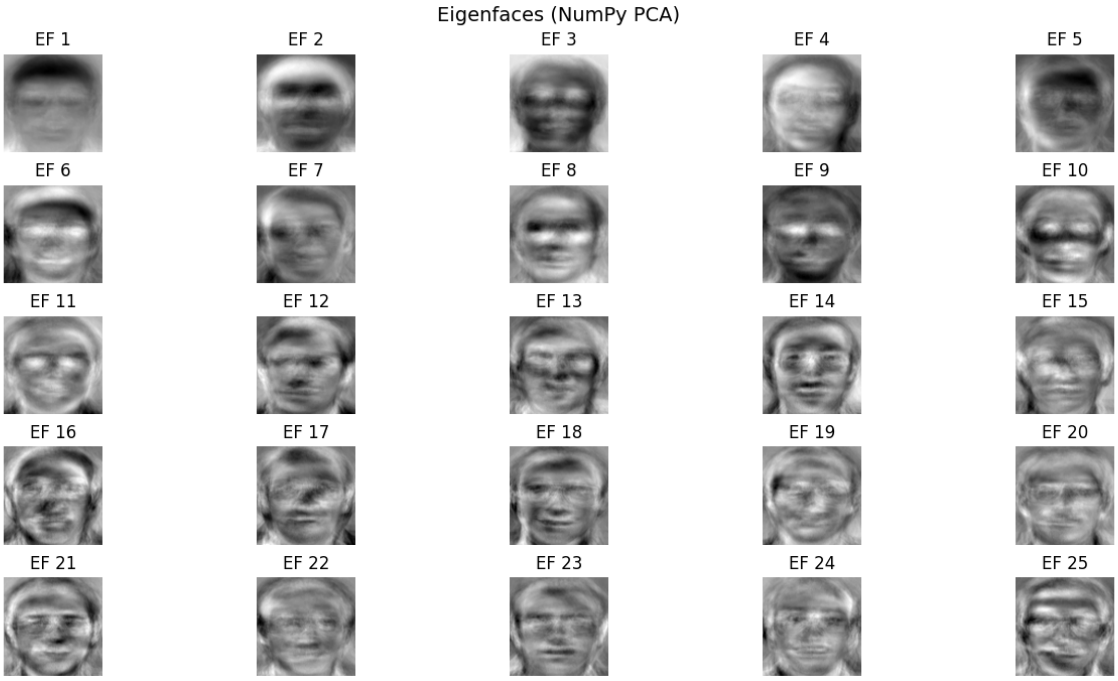


Figure 2. The 25 retained eigenfaces, ordered by decreasing eigenvalue. Each panel displays a unit-length PCA direction reshaped to 100 by 100 pixels.

5.1 Interpretation of the figures

The mean face summarizes the common spatial arrangement of the eyes, nose, and mouth. Its blurred detail follows from averaging faces with different appearances, expressions, and alignments. This image characterizes the center of the dataset; the eigenfaces characterize variation around that center.

The eigenfaces contain broad light-dark patterns as well as more localized structure. These patterns reflect statistical variation rather than individual identities. A component can mix illumination, facial geometry, and other image differences. The sign of an eigenvector is arbitrary, so reversing its contrast does not change the represented direction. Each panel uses its own grayscale display range; visual contrast across panels should not be interpreted as a direct measure of explained variance.

6 Implementation and Discussion

6.1 Computational organization

The implementation separates the workflow into five functions. load_images constructs the image matrix and subject labels. compute_mean_face calculates the average image. compute_eigenfaces centers the data, solves the sample-space eigenproblem, constructs the pixel-space basis, and normalizes its columns. visualize_results displays the mean face and a fixed five-by-five arrangement of eigenfaces. project_faces computes the reduced representation. The main execution block invokes these operations in order.

The Gram-matrix formulation reduces the size of the matrix submitted to eigendecomposition. A dense float64 matrix of size 10,000 by 10,000 requires approximately 762.94 MiB, whereas a 400 by 400 matrix requires approximately 1.22 MiB. The latter contains 625 times fewer entries. These figures describe storage for the individual matrices and exclude other arrays and temporary allocations. No runtime speedup was benchmarked.

6.2 Scope and limitations

PCA is unsupervised in this implementation. Subject labels are stored but do not influence the basis. Consequently, directions of high variance are not guaranteed to be the directions that best distinguish identities. The script does not train a recognition model, predict labels, or compute classification accuracy. Quantitative results therefore concern representation and reconstruction only.

All images contribute to the fitted mean and basis. Evaluation on a separate test set would be necessary to assess representation quality beyond these observations. A recognition experiment would additionally require an explicit classifier and a partition designed to avoid information leakage. Illumination changes, pose variation, and spatial misalignment may influence the basis because they directly affect pixel intensities.

The code assumes that a valid dataset is present and does not explicitly handle an empty dataset, unreadable files, or a requested component count that exceeds the usable numerical rank. The visualizer assumes exactly 25 displayed components. Resizing also changes the original aspect ratio. These assumptions are acceptable for the supplied demonstration but should be addressed before extending the implementation to varied inputs.

6.3 Reproducibility

The numerical analysis used Python 3.12.14, NumPy 2.3.5, and Pillow 12.3.0. The original visualization routine additionally requires Matplotlib. Running the script from the repository root with the dataset under data/ reproduces the loading and PCA workflow. The two figures shown here are the existing project outputs. Eigenvector signs and small floating-point differences can vary across numerical environments without changing the principal subspace.

7 Conclusion

The project implements the central Eigenfaces workflow using explicit NumPy operations. A sample-space eigendecomposition yields 25 numerically orthonormal eigenfaces and reduces each processed image to 25 coefficients. The retained basis captures 74.03% of the centered variance, with a fitted-data reconstruction RMSE of 19.92 intensity units. The numerical diagnostics and visual outputs provide complementary evidence that the implemented operations behave as expected. The resulting representation is a practical basis for subsequent reconstruction and recognition experiments, whose performance remains to be evaluated separately.